



Applications of First Order DE

continuation...

$$v = \frac{ds}{dt}$$

where:

a = acceleration of the body

v = velocity of the body

s = displacement of the body

Also, the Newton's second law of motion can be used in dealing with these types of problems.

$$F = m \frac{dV}{dt}$$

where:

Excel Review Center

F = force

m = mass of the body

$\frac{dV}{dt}$ = rate of change of velocity

7. Hooke's Law and Springs

Hooke's Law of Springs states that the force produced by a spring is proportional to its displacement within limits.

$$F = kx$$

where: k = spring constant

8. Geometrical problems (Orthogonal trajectory)

A curve which intersects all curves of a given family at the same angle is referred to as a **trajectory**; if the intersection is at right angle, the curve is called **orthogonal trajectory**.

$$\left(\frac{dy}{dx}\right)_t = -\left(\frac{dx}{dy}\right)_g$$

Excel Review Center

where:

$\left(\frac{dy}{dx}\right)_t$ = slope of the orthogonal trajectory

$\left(\frac{dy}{dx}\right)_g$ = slope of the given family of curves

$$\left(\frac{dx}{dy}\right)_g = \frac{1}{\left(\frac{dy}{dx}\right)_g}$$

NOTES

NOTES